Monte Carlo method

A model for a stock S is given by the stochastic differential equation

$$dS_t = rS_t\,dt + \alpha S_0^\beta S_t e^{\gamma \sin \delta t}\,dW_t^Q, \tag{1}$$

where W^Q is a Brownian motion under the risk-neutral probability Q. Here $r = 0.05$, $\alpha = 0.3$, $\beta = -0.2$, $\gamma = 0.05$ and $\delta = 3\pi$ are parameters and $S_0 = 100$ is the initial stock price. The aim of this part of the project is to study a European style derivative where the payoff $f(S_T)$ at time $T = 1$ is only paid if the stock price has not his a barrier $L = 110$, in other words, the payoff at time $T = 1$ is

$$f_L(S, (S_t)_{t\in[0,T]}) = \begin{cases} f(S_T) & \text{if } \max_{t\in[0,T]} S_t < L \\ 0 & \text{if } \max_{t\in[0,T]} S_t \geq L \end{cases}$$

The price of this derivative at time 0 is then $e^{-rT}E^Q(f_L(S, (S_t)_{t\in[0,T]}))$.

Exercise 4.1 (Monte Carlo method: 35% of project mark). Implement a Monte Carlo simulation to estimate the price of this derivative at time 0 under Q, as follows.

1. Write down a statement (step by step) of the Monte Carlo procedure to estimate the price of the derivative by applying the Euler scheme. For full marks, use the following techniques:
   - Transform to $Y_t = \ln S_t$ by means of the Ito formula, and then apply the Euler scheme to the process Y.
   - Antithetic sampling.
   - Control variate. Motivate your choice of control variate.

   First off, we apply the transformation $Y_t = \ln S_t$ to our SDE: let $\frac{\partial}{\partial a} \equiv \partial_a$

$$\partial_t Y_t = 0, \qquad \partial_{S_t} Y_t = \frac{1}{S_t}, \qquad \partial^2_{S_t} Y_t = -\frac{1}{S_t^2}$$

   substituting these into the Ito formula, we get

$$\begin{aligned} \mathrm{d}Y_t &= \partial_t Y_t\,\mathrm{d}t + \partial_{S_t} Y_t\,\mathrm{d}S_t + \frac{1}{2}\partial^2_{S_t} Y_t\,\mathrm{d}S_t\,\mathrm{d}S_t \\ &= \frac{1}{S_t}\,\mathrm{d}S_t - \frac{1}{2S_t^2}\,\mathrm{d}S_t\,\mathrm{d}S_t \\ \text{Substitute Eq 1} &= \frac{1}{S_t}\left(rS_t\,dt + \alpha S_0^\beta S_t e^{\gamma \sin \delta t}\,dW_t^Q\right) - \frac{1}{2S_t^2}\left(rS_t\,dt + \alpha S_0^\beta S_t e^{\gamma \sin \delta t}\,dW_t^Q\right)^2 \end{aligned}$$

   This expands to give

$$\begin{aligned} &= r\,\mathrm{d}t + \alpha S_0^\beta e^{\gamma \sin \delta t}\,\mathrm{d}W_t^Q - \frac{1}{2S_t^2}\Big(r^2 S_t^2\,\mathrm{d}t\,\mathrm{d}t + \\ &\qquad + rS_t^2 \alpha S_0^\beta e^{\gamma \sin \delta t}\,\mathrm{d}t\,\mathrm{d}W_t^Q + rS_t^2 \alpha S_0^\beta e^{\gamma \sin \delta t}\,\mathrm{d}W_t^Q\,\mathrm{d}t + +\alpha^2 S_0^{2\beta} S_t^2 e^{2\gamma \sin \delta t}\mathrm{d}W_t^Q\,\mathrm{d}W_t^Q\Big) \end{aligned}$$

   Which simplifies to

$$\begin{aligned} &= r\,\mathrm{d}t + \alpha S_0^\beta e^{\gamma \sin \delta t}\,\mathrm{d}W_t^Q - \frac{1}{2}r^2\,\mathrm{d}t\,\mathrm{d}t - \frac{1}{2}r\alpha S_0^\beta e^{\gamma \sin \delta t}\,\mathrm{d}t\,\mathrm{d}W_t^Q - \\ &\qquad - \frac{1}{2}r\alpha S_0^\beta e^{\gamma \sin \delta t}\,\mathrm{d}W_t^Q\,\mathrm{d}t - \frac{1}{2}\alpha^2 S_0^{2\beta} e^{2\gamma \sin \delta t}\mathrm{d}W_t^Q\,\mathrm{d}W_t^Q \end{aligned}$$

   Now we want to refer to the Ito multiplication table: Using this table simplifies our formula further

	$\mathrm{d}t$	$\mathrm{d}W_t^Q$
$\mathrm{d}t$	0	0
$\mathrm{d}W_t^Q$	0	$\mathrm{d}t$

Table 1: Ito Multiplication table

to give us:

$$\mathrm{d}Y_t = \left(r - \frac{1}{2}\alpha^2 S_0^{2\beta} e^{2\gamma \sin \delta t} \right) \mathrm{d}t + \alpha S_0^{\beta} e^{\gamma \sin \delta t} \, \mathrm{d}W_t^Q$$

Which we note no longer depends on the random process S_t. We can write this as

$$\mathrm{d}Y_t = p_t \, \mathrm{d}t + q_t \, \mathrm{d}W_t^Q \tag{2}$$

where

$$p_t = r - \frac{1}{2}\alpha^2 S_0^{2\beta} e^{2\gamma \sin \delta t}, \qquad q_t = \alpha S_0^{\beta} e^{\gamma \sin \delta t}$$

both only depend on time. Now we can determine a pricing procedure using both antithetic sampling and a control variate:

- Input:
  - Question Parameters: r, α, β, γ, δ, S_0, T
  - Payoff Function $f(S_T)$ and its arguments (K, L, Call or Put)
  - Number of steps M and Number of paths N
- Parameters:
  - Time step: $\Delta t = T/M$
  - Control Variate: $C^* = S_0 e^{rT}$
- Output:
  - Initial Option Price Estimate: η
  - Standard Error SE
  - Confidence interval

(a) For $n = 0, \ldots, N$ do the following:

i. Generate $Z_n \sim N(0,1)$ and calculate $\Delta W_n = \sqrt{\Delta t} Z_n$

ii. Set $Y_{0,n} = \hat{Y}_{0,n} = \ln S_0$ as our initial stock price

iii. For $m = 1, \ldots, M$, do the following:

A. Calculate the transformed estimates (and antithetic pair):

$$Y_{m,n} = Y_{m-1,n} + p\left((m-1)\Delta t\right) \cdot \Delta t + q\left((m-1)\Delta t\right) \cdot \Delta W_n$$
$$\hat{Y}_{m,n} = \hat{Y}_{m-1,n} + p\left((m-1)\Delta t\right) \cdot \Delta t - q\left((m-1)\Delta t\right) \cdot \Delta W_n$$

where $p(t)$ and $q(t)$ are previously defined.

B. Apply the inverse transformation to our estimates to get the stock price estimates:

$$S_{m,n} = e^{Y_{m,n}} \qquad \hat{S}_{m,n} = e^{\hat{Y}_{m,n}}$$

iv. Calculate the payoff of our stocks:

$$P_n = \mathrm{Bar}(\mathbf{S}_n, K, L) \qquad \hat{P}_n = \mathrm{Bar}(\hat{\mathbf{S}}_n, K, L)$$

where $\mathbf{S}_n = [S_{0,n}, S_{1,n}, \ldots, S_{M,n}]$, K is the strike price, L is the barrier price and Bar is our barrier function defined above, which outputs $f(S_{M,n})$ if $\max_{m\in[0,M]} S_{m,n} < L$ and 0 otherwise for any $S_{m,n} \in \mathbf{S}_n$ and similarly for $\hat{\mathbf{S}}_n$.

v. Calculate the antithetic estimate of our terminal stock payoff:

$$X_n = \frac{1}{2}\left(P_n + \hat{P}_n\right)$$

vi. Calculate our control parameter for use with our control variate:

$$C_n = \frac{1}{2}\left(S_{M,n} + \hat{S}_{M,n}\right)$$

(b) and calculate the optimal control variate coefficient b:

$$b = \frac{\mathrm{Cov}(\mathbf{X}, \mathbf{C})}{\mathrm{Var}(\mathbf{C})} = \frac{\sum_{n=1}^{N}(X_n - \bar{X})(C_n - \bar{C})}{\sum_{n=1}^{N}(C_n - \bar{C})^2}$$

where $\bar{X} = \frac{1}{N}\sum_{n=1}^{N} X_n$ and $\bar{C} = \frac{1}{N}\sum_{n=1}^{N} C_n$. We choose to calculate the optimal control variate coefficient b as it minimises the variance of the control variate estimate η. It is estimated from the Monte Carlo sample and converges to the true optimal coefficient as $N \to \infty$!

(c) Calculate control variate estimate of terminal stock price:

$$X_n^{\mathrm{CV}} = X_n - b(C_n - C^*)$$

where C^* is our control variate.

(d) Average over the number of paths N and discount to get the initial option price

$$\eta = e^{-rT}\frac{1}{N}\sum_{n=1}^{N} X_n^{\mathrm{CV}}$$

(e) As an additional step, calculate the discounted standard error of the estimate:

$$SE = e^{-rT}\sqrt{\frac{1}{N(N-1)}\left(\sum_{n=1}^{N}\left(X_n^{\mathrm{CV}} - \bar{X}^{\mathrm{CV}}\right)^2\right)}$$

where $\bar{X}^{\mathrm{CV}} = \frac{1}{N}\sum_{n=1}^{N} X_n^{\mathrm{CV}}$. It is discounted as to represent the standard error of the initial option price as opposed to its terminal price.

4. Write a few paragraphs (no more than 300 words) to discuss whether and how trees and finite difference methods may be applied to approximate the price of the option in the previous task. Which of the three methods (Monte Carlo, tree, finite difference) would you recommend, and why?

   This problem is to price a path-dependent barrier option based on stock that evolves with a time-dependent volatility. Tree methods, robust and efficient at calculating option prices at a current spot rather than over all values, generally lose their effectiveness with path-dependence. This is because recombination is lost when volatility $\sigma(t)$ is time-dependent, meaning a binomial tree method would grow exponentially with $O(2^M)$ nodes, making the general method become very computationally expensive. Finite difference methods are able to price some path dependent options but require us to construct boundary conditions for our problem. The time-dependent volatility means the method would need to recalculate the coefficients of the PDE for each increment in time. As this method must calculate the option price on the full domain of coordinates creating computational cost, a finite difference method could be applied here but requires the careful calculation of the discrete approximation. Monte Carlo methods are designed for a problem of this type, and make use of stochastic processes to estimate integrals and expected values. Using this method, the time dependent volatility would only need to be calculated once. The Ito transformation removes the dependence on the process S_t, converting the problem into a simpler process, improving accuracy showing the Euler scheme that we used is well suited for this problem. Additionally, this method includes many techniques that can be used to reduce the error between the estimate and the true value without increasing the computational cost such as using antithetic samples and or a control variate, much like we applied above. Both the tree and finite difference methods come with unwanted additional computational cost, while the Monte Carlo methods do not and it is for this reason that I would recommend Monte Carlo for a problem of this form.

2. Use your code to estimate the probability that the value of S is equal to or greater than L' at any point in the interval $[0, T]$, where $L' = 100, 101, \ldots, 120$. Display your results in both tabular and graphical format, with an approximate 95% confidence interval for each value of L'. Use 1000 steps per path, and choose the number of samples for each L' such that the confidence interval is at most 0.01 in length.

   See ‘Part_4.ipynb‘ for graphical plot and code. See next page for tabular results.

L'	$P(\max_{t\in[0,T]} S_t \geq L')$	Confidence interval	Interval length
100	1.0000	1.0000 - 1.0000	0.0000
101	0.9492	0.9470 - 0.9513	0.0043
102	0.9052	0.9023 - 0.9080	0.0057
103	0.8607	0.8573 - 0.8640	0.0068
104	0.8138	0.8100 - 0.8176	0.0076
105	0.7663	0.7621 - 0.7704	0.0083
106	0.7220	0.7176 - 0.7264	0.0088
107	0.6772	0.6726 - 0.6818	0.0092
108	0.6317	0.6269 - 0.6364	0.0095
109	0.5886	0.5838 - 0.5934	0.0096
110	0.5434	0.5385 - 0.5483	0.0098
111	0.5021	0.4972 - 0.5070	0.0098
112	0.4631	0.4582 - 0.4680	0.0098
113	0.4248	0.4199 - 0.4296	0.0097
114	0.3874	0.3826 - 0.3921	0.0095
115	0.3525	0.3478 - 0.3571	0.0094
116	0.3189	0.3144 - 0.3235	0.0091
117	0.2882	0.2838 - 0.2927	0.0089
118	0.2597	0.2554 - 0.2640	0.0086
119	0.2332	0.2290 - 0.2373	0.0083
120	0.2094	0.2054 - 0.2134	0.0080

Table 2: Estimated probabilities $P(\max_{t\in[0,T]} S_t \geq L')$ with 95% confidence intervals.